Topology

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Throughout this outline, S is a set of points in which there is determined a collection of subsets called *neighborhoods*, satisfying the following two axioms:

Axiom 1: Each point of S is contained in a neighborhood.

Axiom 2: If A is a neighborhood of a point p, and B is a neighborhood of p, then there exists a neighborhood C of p such that $A \cap B$ contains C.

Definition 3: S is called a *topological space* (or simply a *space*).

Definition 4: A is a *neighborhood* of p if A is a neighborhood and $p \in A$

Definition 5: A point p is an *interior point* of a set H if there exists a neighborhood A of p such that $A \in H$.

Definition 6: A point set H is *open* if every point of H is an interior point.

Proposition 7: Neighborhoods are open.

Proposition 8: The union of any collection of open sets is open.

Proposition 9: The intersection of two open sets is open.

Proposition 10: S is open.

Definition 11: A subset H of S is *closed* if $S - H$ is open.

Proposition 12: The union of two closed sets is closed.

Proposition 13: The intersection of two closed sets is closed.

Question 14: Do Propositions 3, 5, and 6 extend to arbitrary collections?

Definition 15: A point p is a *limit point* of H if every open set containing p contains a point of H distinct from p.

Proposition 16: A point set is closed if and only if it contains each limit point of the set.

Proposition 17: S is closed.

Proposition 18: The empty set is both open and closed.

Proposition 19: A point p is a limit point of H if and only if each neighborhood of p contains a point of H distinct from p.

Definition 20: The *closure* $\overline{H}$ of a point set H, is the union of H and the set of all limit points of H.

Proposition 21: $\overline{H}$ is the intersection of all closed supersets of H.

Corollary 21.1: $\overline{H}$ is closed.

Definition 22: The *interior* (Int H) of a point set H is the set of interior points of H.

Proposition 23: Int H is the union of all open subsets of H.

Definition 24: The *boundary* (Bd H) of H is $\overline{H} -$ Int H.

Proposition 25: Bd $H = \overline{H} \cap \overline{S - H}$

Proposition 26: Bd H is empty if and only if H is both open and closed.

Definition 27: Two non-empty point sets A and B are *separated* if $A \cap \overline{B} = B \cap \overline{A} = \emptyset$.

Definition 28: A point set is *connected* if it is not the union of two separated sets.

Proposition 29: The unit interval $[0,1]$ (with the standard topology) is connected. (Assume every nonempty subset of $[0,1]$ has a least upper bound.)

Proposition 30: S is not connected if and only if there exists a nonempty proper subset H of S such that Bd $H = \emptyset$.

Proposition 31: If H is connected, then so is $H \cup H'$ where H' is in $\overline{H} - H$.

Proposition 32: If each of H and K is connected and $H \cap K \neq \emptyset$, then $H \cup K$ and $H \cap K$ are connected.

Definition 33: A collection G of points sets is *coherent* if G is not the union of two nonempty subcollections E and F such that $E^* \cap F^* = \emptyset$. ($E^* = \cup\{H : H \in E\}$)

Proposition 34: If G is a coherent collection of connected sets, then G^* is connected.

Proposition 35: If C is a connected set and G is a collection of connected sets no member of which is separated from C, then $C \cup G^*$ is connected.

Corollary 35.1: If C is connected and G is a collection of connected sets such that $C \cap H \neq \emptyset$ for each element H of G, then $C \cup G^*$ is connected.

Corollary 35.2: R^n (Euclidean n-space) is connected.

Definition 36: A collection G of open sets is an *open covering* of a set H if H is a subset of G^*.

Definition 37: A set H is *bicompact* if each open covering of H has a finite subcovering.

Proposition 38: The unit interval $[0,1]$ (with the standard topology) is bicompact.

Proposition 39: A closed subset of a bicompact space is bicompact. The union of a finite family of bicompact subsets of a space is bicompact.

Corollary 39.1: The bicompact subsets of R^1 are precisely the closed sets with finite diameters.

Definition 40: A family of sets has the *finite intersection property* (f.i.p) provided each of its finite subfamilies has nonempty intersection.

Proposition 41: A space is bicompact if and only if each family of closed sets having the f.i.p. has nonempty intersection.

Proposition 42: A subset of R^n is bicompact if and only if it is closed and has finite diameter.

Definition 43: A topological space is *regular* if for each point p and each neighborhood N of p, there exists a neighborhood M of p such that $\overline{M}$ is a subset of N.

Proposition 44: S is regular if and only if for each point p and each closed set H not containing p, there exist open sets V and W such that $H \subset V \subset S - W \subset S - \{p\}$.

Definition 45: A point set K is *nowhere dense* if $\overline{K}$ does not contain a nonempty open set.

Proposition 46: No bicompact regular space is the union of countably many nowhere dense sets.

Definition 47: S is a *Hausdorff space* if for each pair x, y of distinct points of S, there exist disjoint open sets X and Y such that x belongs to X and y belongs to Y.

Proposition 48: In a Hausdorff space, a set is bicompact if and only if it is closed.

Proposition 49: A space is Hausdorff if and only if it is regular.

Definition 50: S is *normal* if for each pair E, F of closed disjoint sets, there exist open sets V and W such that $E \subset V \subset S - W \subset S - F$.

Proposition 51: Every bicompact Hausdorff space is regular.

Proposition 52: Ever regular bicompact space is normal.

Definition 53: A space is *Lindelof* if each open covering of the space has a countable subcovering.

Proposition 54: Every Lindelof space is normal.

Definition 55: A *component* of a point set X is a connected subset of X that is not a proper subset of a connected subset of X.

Proposition 56: Each point of a set H belongs to one and only one component of H.

Proposition 57: Is each component of an open set open?

Definition 58: The *p-component* of a set X is the component of X that contains p.

Proposition 59: The p-component of a point set X is the union of all connected subsets of X that contain p.

Proposition 60: If H and K are connected sets, $K \subset H$, and $H - K = A \cup B$ where A, B are separated sets, then $A \cup K$ is connected.

Question 61: Is $K \cup C$ connected when C is a component of $H - K$?

Definition 62: A *continuum* is a bicompact connected Hausdorff space.

Proposition 63: Every nondegenerate continuum is uncountable.

Proposition 64: Every nondegenerate connected regular Hausdorff space is uncountable.

Question 65: Does there exist a nondegenerate countable connected Hausdorff space?

Proposition 66: No continuum is the union of a countable (> 1) family of nonempty disjoint closed sets.

Proposition 67: Suppose p and q are distinct points of a bicompact Hausdorff space M, and suppose $\{H_a : a \in A\}$ is an indexed collection of closed sets in M such that for each $a \in A$, $H_a \subset H_{a+1}$, and H_a cannot be separated between p and q. Then $\cap\{H_a : a \in A\}$ cannot be separated between p and q.

Definition 68: A continuum C is *irreducible between two disjoint sets* if C intersects each set and no proper subcontinuum of C intersects both sets.

Proposition 69: Each pair of points of a continuum lies in a subcontinuum irreducible between the two points.

Definition 70: A continuum H is *irreducible about a set* K if H contains K and no proper subcontinuum of H contains K.

Proposition 71: If K is any subset of a continuum C, then C contains a subcontinuum irreducible about K.

Proposition 72: Suppose X is a bicompact Hausdorff space, and p and q belong to different components of X. Then there exist separated sets H and K such that p belongs to H, q belongs to K, and $X = H \cup K$.

Question 73: Does Proposition 72 hold when X is not bicompact?

Proposition 74: Suppose C is a continuum, and D is an open proper subset of C that contains a point p. Then Bd D contains a limit point of the p-component of D.

Definition 75: Suppose that $\{X_n\}$ is a sequence of subsets of S. The set of all points x in S such that every open set containing x intersects all but a finite number of elements of $\{X_n\}$ is called the *limit inferior* of $\{X_n\}$ (lim inf X_n). The set of all points y in S such that every open set containing y intersects infinitely many elements of $\{X_n\}$ is called the *limit superior* of $\{X_n\}$ (lim sup X_n).

Proposition 76: lim inf $X_n =$ lim sup X_n.

Proposition 77: lim sup $X_n \neq \emptyset$.

Proposition 78: If lim inf $X_n \neq \emptyset$, then lim sup $X_n \neq \emptyset$.

Proposition 79: lim inf $X_n =$ lim inf $\overline{X_n}$ and lim sup $X_n =$ lim sup $\overline{X_n}$.

Proposition 80: lim inf X_n and lim sup X_n are both closed sets.

Proposition 81: If $\{X_n\}$ is a sequence of connected sets in a continuum, and if lim inf X_n is not empty, then lim sup X_n is a continuum.

Definition 82: A function f of S into a space T is *continuous* provided that if p is a point of the closure of a subset X of S, then $f(p)$ belongs to $\overline{f[x]}$. A continuous function is sometimes called a *map*.

Proposition 83: For real-valued functions of one real variable, this definition is equivalent to the standard $\delta - \epsilon$ definition.

Proposition 84: Suppose f is a function of S into a space T. Then f is continuous if and only if for each open set G of T, $f^{-1}[G]$ is open in S.

Question 85: Does Proposition 84 hold when the word "open" is everywhere replaced by the word "closed"?

Proposition 86: Every continuous image of a connected space is connected.

Proposition 87: Every continuous image of a bicompact space is connected.

Question 88: Is the Lindelof property a continuous invariant?

Proposition 89: Every continuous image of a Hausdorff space is Hausdorff.

Proposition 90: The composition of two continuous functions is continuous.

Proposition 91: Suppose X is a bicompact space, Y is a Hausdorff space, and f is a one-to-one map of X onto Y. Then f^-1 is continuous.

Question 92: Does Proposition 91 hold when X is not required to be bicompact?

Proposition 93: Every line interval $[a, b]$ is an image of $[0, 1]$ under a one-to-one map.

Proposition 94: S is a normal space if and only if for each pair A, B of disjoint closed subsets of S, there exists a map f of S into a line interval $[a, b]$ such that $f[A] = a$ and $f[B] = b$.

Proposition 95: For each positive integer n, let f_n be a map of S into the real line. Suppose there exists a convergent series of positive numbers $\sum_{n=1}^{\infty}$ M_n, such that for each point x of S and each n, $|f_n(x)| \leq M_N$. Then for each point x of S, the infinite series $\sum_{n=1}^{\infty} f_n(x)$ converges to a number $f(x)$, and the function f so defined is continuous.

Definition 96: S has the *continuous extension property* provided that for each map f of any closed subset H of S into an interval $[a, b]$, there exists a map f^* of S into $[a, b]$ such that $f^*(x) = f(x)$ for each point x of H.

Proposition 97: S is normal if and only if S has the continuous extension property.

Proposition 98: Suppose S is normal, and f is a map of a closed subset H of S into I^n (the Cartesian product of $[0, 1]$ with itself n times). Then there exists an extension f^* of f to all of S (i.e., $f^* : S \to I^n$ and $f^*(x) = f(x)$ for each point x of H).

Definition 99: A collection C of open sets is a *base* for S if for each point p of S, and each open set U of S that contains p, there exists an element G of C such that $p \in G \subset U$.

Definition 100: The *Cartesian product* of the indexed collection of sets $\{X_a : a \in A\}$ is the set of functions $\times\{X_a : a \in A\} = \{f : A \to \cup\{X_a : a \in A\}$ and $f(a) \in X_a\}$.

Definition 101: The a-th *projection map* is the function $P_a : \times\{X_a : a \in A\} \to X_a$ such that $P_a(x) = x_a$ for every point x of $\times\{X_a : a \in A\}$.

To define the standard topology given to the Cartesian product, called the *product topology*, we define the base.

Definition 102: A subset D of $\times\{X_a : a \in A\}$ belongs to D if there is a finite subset $\{a_1, a_2, \cdots, a_n\}$ of A, and open subsets $D_1, D_2, \cdots, D_n$ of $X_{a_1}, X_{a_2}, \cdots, X_{a_n}$, respectively, such that $D = \{x \in \times\{X_a : a \in A\} : x_{a_1}$ belongs to D_i for $i = 1, 2, \cdots, n\}$. The collection forms a base for the product topology.

Proposition 103: If D is open in $\times\{X_a : a \in A\}$, then $P_a[D]$ is open for every a belonging to A.

Proposition 104: A function $f : S \to \times\{X_a : a \in A\}$ is continuous if and only if $P_a f$ is continuous for each a belonging to A.

Proposition 105: The space $\times\{X_a : a \in A\}$ is Hausdorff if and only if each X_a is a Hausdorff space.

Proposition 106: The product of two connected spaces is connected.

Proposition 107: The product of two normal spaces is normal.

Proposition 108: The product of two bicompact spaces is bicompact.

Question 109: Do Propositions 106, 107, and 108 extend to all Cartesian products?

Definition 110: A function $d : S \times S \to [0, +\infty)$ is a *metric* for S if for every x, y, and z belonging to S:

1. $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, y) = d(y, x)$; and
3. $d(x, y) + d(y, z) \geq d(x, z)$.

Definition 111: S is a *metrizable space* (*metric space*) if there exists a metric d for S such that the collection $\{N(x, r) : x$ belongs to S and $r > 0\}$ is a base for the topology of S where $N(x, r) = \{y : d(x, y) < r\}$.

Proposition 112: The function d (defined above) is continuous.

Definition 113: S is *first countable* if for each point p of S there exists a countable collection C of open sets (each containing p) such that every open set in S that contains p also contains an element of C.

Proposition 114: S is metrizable if and only if S is first countable.

Definition 115: A set is said to be of *type* G_δ if it is the intersection of countably many open sets.

Proposition 116: Every closed subset of a metric space is of type G_δ.

Proposition 117: S is *second countable* if S has a countable base.

Proposition 118: Every bicompact metric space is second countable.

Definition 119: A sequence $\{X_n\}$ is a convergent sequence if lim inf $X_n =$ lim sup X_n.

Proposition 120: If M is a bicompact metric space, then every sequence of subsets of M has a convergent subsequence.

Definition 121: A set D is a *dense subset* of a set E if $D \subset E \subset \overline{D}$.

Definition 122: S is a *separable* space if there exists a countable dense subset of S.

Proposition 123: A metric space is separable if and only if it is second countable.

Proposition 124: Proposition 123 holds for all spaces.

Proposition 125: Proposition 120 holds for separable metric spaces.

Proposition 126: A metric space M is bicompact if and only if every infinite subset of M has a limit point.

Proposition 127: Proposition 126 holds for all spaces.

Proposition 128: If every uncountable subset of a metric space M has a limit point, then every uncountable subset of M has a limit point.

Proposition 129: A metric space M is Lindelof if and only if every uncountable subset of M has a limit point.

Proposition 130: Proposition 129 holds for all spaces.

Definition 131: A sequence of points p_i in a metric space (M, d) is a *Cauchy sequence* if for each real number $r > 0$, there exists an integer N such that $m, n > N$ implies $d(p_m, p_n) < r$. The metric space (M, d) is *complete* if each Cauchy sequence in M has a limit point in M.

Definition 132: A *homeomorphism* is a one-to-one map whose inverse is continuous.

Proposition 133: Suppose (M_1, d_1) and (M_2, d_2) are metric spaces, and there exists a homeomorphism of M_1 onto M_2 (i.e., M_1 and M_2 are homeomorphic). Then (M_2, d_2) is complete if (M_1, d_1) is complete.

Proposition 134: Every closed subspace of a complete metric space is a complete metric space (with respect to the restricted metric).

Proposition 135: Suppose D is a dense type G_δ subset of a complete metric space (M, d). Then D is not the union of countably many nowhere-dense subsets of M.

Question 136: Is D in Proposition 135 a complete metric space?

Proposition 137: The rationals are not a G_δ subset of R^1.

Definition 138: Let $\#$ denote the first uncountable ordinal, and let $[1, \#]$ denote the set of all ordinals up to and including $\#$ (similarly, $[1, \#)$ denotes those up to but not including $\#$). For every pair of ordinals a and b $(a < b)$ in $(1, \#)$, the segment $(a, b) = \{x : a < x < b\}$ and the sects $[1, a)$ and $(a, \#]$ are neighborhoods.

Proposition 139: If C is a collection of neighborhoods that cover $[1, \#)$, then there exists $x < \#$ such that $\cup\{G \in C : x \in G\}$ covers $[x, \#)$.

Proposition 140: Suppose f is map of $[1, \#)$ into R^1. Then a point x of $[1, \#)$ and a real number r exist such that $f[[x, \#)] = r$.

Proposition 141: $[1, \#)$ and $[1, \#]$ are normal.

Proposition 142: $[1, \#]$ is metrizable.

Proposition 143: $[1, \#)$ is metrizable.

Proposition 144: $[1, \#]$ is bicompact.

Definition 145: A metric space S is **complete** if there exists a metric k for S such that (S, k) is complete.

Proposition 146: $[1, \#)$ is bicompact.

Proposition 147: Every infinite subset of $[1, \#)$ has a limit point.

Proposition 148: The product of countably many metric spaces is metrizable.

Proposition 149: The product obtained by crossing $[0, 1]$ with itself countably many times is metrizable.

Proposition 150: Proposition 148 is true if the word "countable" is replaced by the word "uncountable".

Proposition 151: Every regular second countable Hausdorff space is metrizable.